# Chapter 2

## Capacity Utilization in the Center and the Periphery: Old Debates and New Estimates

Dissertation Chapter — Canonical Scaffold

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**Abstract**

**[DRAFT PROSE: Abstract: 150–200 words. State: (1) the structural deficiency of existing CU estimates and why it matters for crisis theory; (2) the RRR/CVAR identification strategy and what it recovers; (3) the behavioral recapitalization law and its nested closures; (4) the Tavares peripheral twist as the chapter's distinctive contribution; (5) the three downstream objects inherited by Ch3.]**

# 1 Introduction

Capacity utilization occupies a peculiar position in macroeconomic analysis: universally invoked, rarely identified. The profit rate decomposes as $r_t = \mu_t \cdot B_t \cdot \pi_t$ — utilization scales normal profitability — so the path of $\mu_t$ mediates every link between accumulation dynamics and crisis. Yet the capacity utilization series that enter empirical work are typically constructed by statistical filters that impose balanced growth by assumption, setting the capacity transformation elasticity $\theta$ to unity and suppressing the regime-dependent content through which structural disequilibrium operates. Chapter 1 demonstrated this deficiency: existing CU measures — from the CBO output gap to the Banco Central de Chile's structural balance framework — recover a mean-reverting residual by construction, not a structurally identified object. This chapter asks a different question: *what does capacity utilization do in the economy, and how does its role differ between center and periphery?*

The answer proceeds through a two-stage empirical machine applied identically to the United States and Chile. Stage A recovers capacity utilization as a latent confinement object. The structural primitives are accounting identities: output decomposes into productive capacity and utilization ($y_t = y_t^p + \ln \mu_t$), productive capacity is linked to the capital stock through the transformation elasticity ($y_t^p = \theta(\Lambda)\, k_t$), and the confinement identity $\ln \mu_t = y_t - \theta(\Lambda)\, k_t$ follows immediately. The parameter $\theta(\Lambda)$ is an upstream regime parameter pinned by the institutional space — not a behavioral choice and not derivable from any first-order condition. The empirical specification embeds distribution through the interaction term $\pi_t k_t$, yielding the long-run capacity relation (**??**) with endogenous elasticity $\theta_t = \beta_1 + \beta_2 \pi_t$. If the residual is stationary, the recovered $\widehat{\mu}_t$ is the structurally identified utilization series — and with it come the derived objects $\hat{B}_t$ (capital productivity at normal capacity) and $\hat{y}_t^p$ (productive capacity) that Stage B requires as inputs.

Stage B estimates a behavioral accumulation law. The accounting identity $\iota_t = \beta_t\, r_t$ decomposes the gross accumulation rate into the recapitalization rate and the profit rate; the behavioral question is whether $\beta_t$ responds to distribution, demand, and the structural

transformation of capital productivity. The behavioral accumulation law (**??**) nests three closures — Cambridge, Bhaduri–Marglin, and full structural — testable via Wald restrictions on the coefficients $b_j = 1 + \eta_j$. The deviation from unity is the behavioral content.

The same machine is run on both countries, and the structural comparison is meaningful because the measurement apparatus is identical. The key asymmetry that emerges is not a measurement artefact but a structural finding: in the periphery, the recapitalization response to capital productivity is constrained by the availability of foreign exchange. The Tavares channel — the chapter's distinctive theoretical contribution — localizes this constraint in the recapitalization function rather than in the output equation. Peripheral accumulation requires imported machinery; the capital-goods sector is structurally incomplete; and $\beta_t$ collapses when the foreign currency needed to purchase imported capital goods is unavailable (**?**). The coefficient $b_3$ — the recapitalization response to capital productivity — is the empirical locus of this constraint, and the comparison $b_3^{US}$ versus $b_3^{CL}$ gives the Tavares channel testable content.

The remainder of the chapter is organized as follows. Section **??** establishes the crisis-theoretic motivation. Section **??** develops the accounting scaffold. Section **??** introduces the distributive micro-foundation. Section **??** bridges the Ch1 focal data pair to Ch2's state vector. Section **??** estimates the structural identification (Stage A). Section **??** derives the accounting law of capital accumulation. Section **??** estimates the behavioral recapitalization law (Stage B). Section **??** formalizes the peripheral twist. Section **??** synthesizes the center–periphery comparison. Section **??** concludes.

# 2 Crisis Theory: Stagnation Tendencies, Cumulative Disequilibrium, and the Role of Capacity Utilization

Chapter 1 established that existing capacity utilization measures are structurally deficient: they impose balanced growth by construction and suppress the regime-dependent content

of the capital–output relationship. This section establishes why that deficiency matters. Capacity utilization is the mechanism through which accumulation dynamics trigger crisis — and without a structurally identified $\mu_t$, the crisis mechanism cannot be read empirically.

## 2.1 Stagnation and Crisis: A Typology

The empirical analysis of capacity utilization in this chapter is motivated by a prior question: what role does utilization play in the dynamics of capitalist crisis? Before that question can be answered, the varieties of crisis must be distinguished. The stagnation typology developed by **??** and **?** organizes the structural coordinates within which crises unfold and clarifies where capacity utilization enters the causal chain.

The typology is structured as a two-by-two matrix along two axes (Table **??**). The rows distinguish the source of disruption: demand-side mechanisms, in which the realization of surplus value is interrupted, and financial-side mechanisms, in which the circuits of credit and asset valuation that mediate accumulation break down. The columns distinguish the distributional configuration: an excess of surplus value, in which profits are abundant relative to the economy's capacity to absorb them productively, and a deficit of surplus value, in which profits are squeezed by rising labor strength or deteriorating productivity conditions.

Each cell identifies a distinct crisis tendency. Under-consumption arises when the surplus is generated but cannot be realized through effective demand. Profit squeeze describes the configuration in which rising labor strength compresses the profit share, eroding the incentive to accumulate. Financial fragility emerges when abundant surplus is channeled into speculative asset positions rather than productive investment, generating Minsky-type instability. The falling rate of profit — the classical Marxian tendency — operates when capital deepening outpaces productivity gains and the profit rate declines secularly.

These four tendencies are not mutually exclusive; concrete historical episodes typically combine elements of more than one cell. What matters for the present chapter is the distinction between partial and structural crisis. A partial crisis is an interruption resolvable

by partial reconfiguration of the institutional space $\Lambda$ — a policy adjustment, a shift in the terms of class compromise, a financial restructuring. A structural crisis requires drastic restructuring of the accumulation regime itself: a change in the technology of production, the organization of labor markets, the international division of labor, or the financial architecture.

The typology locates the structural coordinates within which crises develop, but it does not specify the mechanism that triggers the transition from accumulation to crisis. That trigger — the cumulative disequilibrium in the path of capacity utilization — is the subject of the next subsection.

Table 1: Stagnation Tendencies: A Typology

| | Excess of Surplus Value | Deficit of Surplus Value |
|---|---|---|
| Demand side | Under-consumption / realization crisis | Profit squeeze (rising labor strength) |
| Financial side | Financial fragility | Falling (normal) rate of profit |

*Notes*: Following **??**. Each cell denotes a qualitatively distinct stagnation tendency. Partial crisis $\equiv$ interruption resolvable by partial reconfiguration of the institutional space $\Lambda$. Structural crisis $\equiv$ deep interruption requiring drastic restructuring.

## 2.2 Okishio's Crisis Trigger and Cumulative Disequilibrium

The stagnation typology of §**??** identifies the structural coordinates within which crises unfold, but it does not specify a trigger. What sets a crisis in motion? **?**'s (**?**) contribution provides the answer that organizes this chapter: decentralized capitalist investment decisions produce a secular cumulative disequilibrium in capacity utilization, and it is the *path* of $\mu_t$ — not a static threshold — that triggers the transition from partial adjustment to structural crisis.

The core of Okishio's argument is that the mismatch between effective demand and productive capacity is not self-correcting. Individual capitalists, responding to current profitability signals, expand capacity in ways that are locally rational but globally destabilizing.

When demand is strong and utilization is high, investment responds — but the resulting capacity additions do not arrive instantaneously, and when they do, the aggregate demand conditions that motivated them may no longer prevail. The consequence is a secular tendency for $\mu_t$ to trace an unstable path: episodes of high utilization attract investment that eventually depresses utilization, while episodes of low utilization discourage investment and allow demand to recover, but each cycle leaves a residue of structural imbalance. Utilization rates trend toward a reversal of normal capacity but overshoot, producing stagnation episodes of increasing severity.

The critical insight is that the crisis trigger is the path of $\mu_t$, not a level. There is no fixed utilization rate at which crisis erupts. Rather, the cumulative trajectory — the sequence of overshoots and undershoots, the widening amplitude of cyclical swings, the secular drift of the utilization path relative to the capacity manifold — is itself the destabilizing mechanism. This is what distinguishes Okishio's framework from static equilibrium models in which utilization gravitates toward a given normal rate: the disequilibrium is cumulative, and its accumulation is the crisis.

The channel through which the utilization path transmits into crisis runs through profitability. The **?** (**?**) decomposition of the profit rate provides the accounting link:

$$r_t = \mu_t \cdot B_t \cdot \pi_t, \tag{1}$$

where $r_t \equiv \Pi_t/K_t$ is the profit rate, $\mu_t \equiv Y_t/Y_t^p$ is capacity utilization, $B_t \equiv Y_t^p/K_t$ is capital productivity at normal capacity, and $\pi_t \equiv \Pi_t/Y_t$ is the profit share. Profitability is demand-led through $\mu_t$: utilization scales normal profitability $B_t\pi_t$ up or down, so that the entire profit rate fluctuates with the utilization path. When $\mu_t$ falls persistently — when the cumulative disequilibrium that Okishio describes tilts toward chronic underutilization — profitability erodes even if $\pi_t$ remains stable and $B_t$ does not deteriorate. The crisis is transmitted from the demand side to the profitability side through the utilization channel,

and from profitability to accumulation through the investment response.

This transmission mechanism makes the identification of $\mu_t$ a precondition for any empirical analysis of the crisis–accumulation nexus. To read (**??**) as an empirical statement rather than a tautology, one needs a structurally identified utilization series — one that is not imposed by a statistical filter but recovered from the long-run reproduction structure of the economy. The existing CU estimates examined in Chapter 1 impose balanced growth by construction ($\theta = 1$), suppressing precisely the regime-dependent content through which cumulative disequilibrium operates. The structural primitives of §**??** provide the alternative: a confinement identity in which $\mu_t$ is recovered as the deviation of realized output from the productive-capacity manifold under unbalanced growth. The next subsection traces the link from profitability to accumulation that completes the crisis mechanism.

## 2.3 From Profitability to Accumulation: The Missing Link

The preceding subsection established that the path of $\mu_t$ transmits cumulative disequilibrium into profitability through the Weisskopf decomposition (**??**). But the crisis mechanism is incomplete without a further link: profitability must translate into accumulation for the disequilibrium to propagate. The literature on the tendency of the profit rate to fall (**??**) has documented the long-run trajectory of $r_t$ in the US and other advanced economies, yet the causal channel from profitability to crisis runs not through the profit rate itself but through the investment response it elicits. As **?** and **??** argue, the rate of profit matters for crisis dynamics because it drives the rate of investment — and it is the rate of investment, not the rate of profit, that determines whether the economy reproduces or contracts its productive capacity.

This chapter formalizes that link through the behavioral accumulation law developed in §**??**–**??**. The gross accumulation rate $\iota_t = I_t/K_t$ decomposes as $\iota_t = \beta_t\, r_t$, where $\beta_t$ is the recapitalization rate. The behavioral question is whether $\beta_t$ responds to the structural conditions summarized by $\pi_t$, $\mu_t$, and $B_t$. The estimation of that response — carried out in

Stage B of the empirical machine — requires structurally identified objects: the utilization series $\widehat{\mu}_t$ and the capital-productivity series $\hat{B}_t$ recovered from Stage A.

The mechanism differs structurally between center and periphery. In the US, the recapitalization rate responds to the three profitability channels with relatively few binding external constraints. In Chile, the foreign-exchange constraint mediates the conversion of surplus into investment goods, so that the recapitalization response to capital productivity — the $b_3$ coefficient — is structurally dampened. The Tavares channel (§**??**) provides the theoretical mechanism for that dampening.

To make the crisis-trigger hypothesis testable, $\mu_t$ must be identified structurally — not imposed by a statistical filter. Chapter 1 showed why existing measures fail; the next section formalizes the alternative.

# 3 Structural Primitives: Production, Capacity, and Unbalanced Growth

This section establishes the accounting scaffold on which all subsequent identification and estimation rests. Every relation derived here is a definition or an identity; no behavioral equation is imposed until §**??**. The reader should treat each step as a manipulation of national-accounts magnitudes, not as a claim about agent optimization.

We adopt a Leontief production structure in which both capital and labor bind at full capacity utilization:

$$Y_t = \min(B_t K_t,\ A_t L_t)\,, \tag{2}$$

with both constraints binding at $\mu_t = 1$. Output decomposes into productive capacity and

utilization:

$$Y_t = \mu_t \cdot Y_t^p, \tag{3}$$

$$y_t = y_t^p + \ln \mu_t, \tag{4}$$

where lowercase denotes natural logarithms of the corresponding uppercase level, $y_t \equiv \ln Y_t$, $k_t \equiv \ln K_t$. Capital productivity at normal capacity is

$$B_t \equiv \frac{Y_t^p}{K_t}, \qquad \ln B_t = y_t^p - k_t. \tag{5}$$

The long-run transformation of capital stock into productive capacity is governed by the *unbalanced growth closure*:

$$y_t^p = \theta(\Lambda) \cdot k_t, \tag{6}$$

where $\theta(\Lambda)$ is the capacity transformation elasticity — a regime parameter, pinned by the institutional space $\Lambda$ and not derivable from within-period optimization. Balanced growth ($\theta(\Lambda) = 1$) is the knife-edge special case; the general case is $\theta(\Lambda) \neq 1$.

**Proposition 3.1** (Confinement Identity)**.** *Given the decomposition* (**??**) *and the unbalanced growth closure* (**??**),

$$\ln \mu_t = y_t - \theta(\Lambda)\, k_t. \tag{7}$$

*Capacity utilization is identified as the deviation of realized output from the productive-capacity manifold.*

The confinement identity is the core identification object of the chapter. Three features merit emphasis. First, it is pure accounting: no behavioral assumption has been invoked, only the decomposition of output into capacity and utilization and the definition of the capacity transformation elasticity. Second, when $\theta(\Lambda) \neq 1$ the identity implies persistent structural disequilibrium in the capital–output ratio — the economy does not settle on a

balanced-growth path, and the deviation of $\theta$ from unity measures the degree of structural imbalance. Under-mechanization ($\theta > 1$) implies that capital accumulation generates more than proportional capacity expansion; over-mechanization ($\theta < 1$) implies that the economy accumulates capital faster than it creates productive capacity, so that the capital–output ratio rises secularly. Third, the Harrodian benchmark $\theta = 1$ is not the default but the special case: imposing it amounts to assuming away precisely the structural content that this chapter seeks to recover.

To take the confinement identity to data, one requires an empirically tractable specification of $\theta$. The next section introduces the distribution-conditioned capacity relation, in which the transformation elasticity varies with the profit share, and derives the rank-1 identification equation that makes $\mu_t$ recoverable from observed time series.

# 4 Distribution-Conditioned Capacity: The Interaction Term

The confinement identity derived in §**??** recovers capacity utilization as the deviation of realized output from the productive-capacity manifold, but it leaves $\theta(\Lambda)$ as an abstract regime parameter. To take the identity to data, one needs an empirically tractable specification that links $\theta$ to observable magnitudes. This section introduces the distribution-conditioned capacity relation, in which the transformation elasticity varies with the profit share, and derives the rank-1 identification equation that makes $\mu_t$ recoverable from observed time series.

Let the exploitation rate $e_t$ map into the profit share as

$$\pi_t = \frac{e_t}{1 + e_t}, \tag{8}$$

and introduce the interaction term $z_t \equiv \pi_t \cdot k_t$ as the empirical proxy for $\theta(e|\Lambda) \cdot k_t$ in the

cointegrating space. The long-run capacity relation then takes the estimable form:

$$y_t = c + \beta_1 k_t + \beta_2(\pi_t k_t) + \xi_t, \qquad \xi_t \equiv \ln \mu_t, \tag{9}$$

so that the endogenous long-run transformation elasticity is

$$\theta_t = \beta_1 + \beta_2 \, \pi_t. \tag{10}$$

**Proposition 4.1** (Rank-1 Identification)**.** *If $\xi_t = y_t - c - \beta_1 k_t - \beta_2(\pi_t k_t)$ is stationary, then*

$$\widehat{\ln \mu_t} = y_t - \hat{c} - \hat{\beta}_1 k_t - \hat{\beta}_2(\pi_t k_t) \tag{11}$$

*is the recovered latent utilization object. This is the dissertation's primary identification problem.*

Three features of this identification deserve emphasis. First, $\theta$ remains upstream: the parameters $\beta_1$ and $\beta_2$ are structural parameters of the cointegrating relation, not behavioral margins subject to agent optimization. The linearization (**??**) is the empirical proxy for the general $\theta(e|\Lambda)$ introduced in §**??**; the capacity transformation elasticity is a property of the institutional regime, not a choice variable. Second, the interaction term $z_t = \pi_t k_t$ rotates the cointegrating space — it is a constructed object needed to represent $\theta(e) \cdot k$ within the linear structure of the Johansen VECM. Without it, the cointegrating relation would impose a constant $\theta = \beta_1$, collapsing the distribution-conditioned specification back to the regime-invariant case. Third, the Increasing Intensity of Conflict Schedule (IICS) is a falsifiable hypothesis, not an imposed condition. The sign of $\beta_2$ is unrestricted: if $\beta_2 > 0$, then higher exploitation raises $\theta_t$ and distributional polarization expands the economy's capacity-generating ability per unit of capital; if $\beta_2 < 0$, the opposite holds; if $\beta_2 = 0$, distribution does not condition the transformation elasticity and the specification reduces to the confinement identity of §**??** with a constant slope. The data adjudicate. The next

section embeds this identification in the center–periphery comparison that structures the empirical work of the chapter.

# 5 From Chapter 1 to Chapter 2: The Focal Data Pair and the State Vector

Chapter 1 established the bivariate output–capital pair as the empirically validated focal data for US capacity utilization, diagnosed rank-1 confinement via ARDL bounds tests, and recovered a preliminary transformation elasticity. This section maps that focal pair into the trivariate cointegrating state vector of Chapter 2, extends the construction to Chile, and states the methodological rationale for the parallel design.

## 5.1 US: From Ch1's ARDL to Ch2's Cointegrating Relation

**[DRAFT PROSE: §2.5.1: 1.5 pages. Ch1's bivariate ARDL on $(y_t, k_t)$ with VAcorp/KGCcorp recovered $\hat{\theta}^{S1}$ and diagnosed rank-1 confinement (S2). Ch2 augments with $z_t = \pi_t k_t$, allowing $\theta_t$ to vary with distribution. State vector: $\mathbf{X}_t^{US} = (y_t,\, k_t,\, \pi_t k_t)'$. Data: BEA/FRED $(y, k)$ from Ch1; NIPA Table 1.12 $(e \to \pi)$. Sample: 1947–2019. Positioning: contrast with CBO, Adams–Coe, multivariate filter methods that impose $\theta = 1$.]**

The state vector for the US system is

$$\mathbf{X}_t^{US} = \begin{pmatrix} y_t \\ k_t \\ \pi_t k_t \end{pmatrix}, \tag{12}$$

estimated on annual data from 1947 to 2019, with output and capital stock inherited from the Chapter 1 BEA/FRED pipeline, and the profit share constructed from NIPA Table 1.12.

## 5.2 Chile: Constructing the Parallel

The Chilean empirical construction mirrors the US case in identification strategy while confronting distinct data challenges and a fundamentally different institutional context for capacity utilization measurement. The data construction requires splicing across multiple statistical vintages. The Chilean capital stock series draws on **?** for the pre-1960 period, the Penn World Table (`rnna`) for the post-1960 benchmark, and the **?** and **?** series for interpolation and cross-validation. The splicing procedure matches growth rates at overlap years rather than levels, preserving the long-run trend properties that the cointegration analysis requires. The profit share is constructed from national accounts data on compensation of employees, spliced with the **?** adjusted wage-share series for the pre-1990 period and cross-checked against the **?** historical distributional series. The exploitation rate $e_t$ is then recovered from $\pi_t = e_t/(1 + e_t)$, and the interaction term $z_t^{CL} = \pi_t^{CL} k_t^{CL}$ is constructed exactly as in the US case. The estimation sample runs from 1960 to 2019, constrained at the lower end by capital-stock availability.

The positioning of this identification against the existing Chilean CU literature is consequential. Capacity utilization in Chile is not merely an academic object — it is a governance variable that feeds directly into the structural fiscal balance rule administered by the Banco Central de Chile and the Dirección de Presupuestos. The BCCh framework, formalized in **?** and updated in **?** and **?**, estimates potential output using multivariate filter methods that impose stationarity of the output gap by construction. The **?** IMF estimates for Chile adopt a similar approach. In all these frameworks, the transformation elasticity $\theta$ is implicitly set to unity: capital and capacity are assumed to grow at the same rate, and the output gap is a mean-reverting deviation from a balanced-growth trend.

The identification developed in this chapter offers an alternative. By permitting $\theta \neq 1$ and recovering the distribution-conditioned capacity slope, it allows the data to reveal whether the Chilean economy operates under over-mechanization ($\theta < 1$) — a possibility that the structural-balance framework rules out by assumption. The key asymmetry between the US

and Chilean cases — that CU estimates in the US serve primarily as academic benchmarks while in Chile they govern fiscal policy directly — is not an artefact of measurement but a finding that emerges from the institutional embedding of the concept.

The state vector for the Chilean system mirrors (**??**):

$$\mathbf{X}_t^{CL} = \begin{pmatrix} y_t^{CL} \\ k_t^{CL} \\ \pi_t^{CL} k_t^{CL} \end{pmatrix}. \tag{13}$$

## 5.3 The Parallel Construction as Method

Both countries receive the same state vector $\mathbf{X}_t = (y_t, k_t, \pi_t k_t)'$, the same Johansen cointegration procedure, and the same rank-1 identification equation derived in §**??**. The structural comparison in §**??** is meaningful because the measurement apparatus is identical: any difference in the estimated $\theta$, in the recovered $\widehat{\mu}_t$, or in the behavioral coefficients $b_1, b_2, b_3$ reflects a structural difference between the two economies, not a difference in how the objects were constructed.

This is the methodological core of the parallel design. The US budget-baseline embedding of CU — in which utilization estimates inform Congressional Budget Office projections but carry no direct fiscal consequence — and the Chilean structural-balance embedding — in which CU estimates govern the fiscal expenditure ceiling — are not parallel by construction. They are asymmetric institutional facts. That asymmetry is a finding of the comparison, not an artefact of the method. The parallel construction isolates the source of difference: because the identification is the same, the differences that emerge in §**??** can be attributed to the structural and institutional conditions under which accumulation proceeds, not to the econometric choices that produced the estimates.

# 6 Stage A: Structural Identification (RRR/CVAR)

Stage A estimates the long-run capacity relation derived in §**??** and recovers the latent utilization object $\widehat{\mu}_t$. The econometric framework is the **?** maximum-likelihood procedure applied to $\mathbf{X}_t = (y_t, k_t, \pi_t k_t)'$, with rank determination by trace and maximum-eigenvalue tests, a restricted constant specification, and **?** endogenous break detection. The estimation setup is common to both countries; results are reported separately below.

## 6.1 Estimation

The **?** procedure embeds the long-run capacity relation within a vector error-correction model (VECM) of the form

$$\Delta \mathbf{X}_t = \Pi \mathbf{X}_{t-1} + \sum_{i=1}^{p-1} \Gamma_i \Delta \mathbf{X}_{t-i} + \Phi D_t + \varepsilon_t, \tag{14}$$

where $\Pi = \alpha\beta'$ is the reduced-rank matrix that encodes the cointegrating relations, $\Gamma_i$ are short-run adjustment matrices, and $D_t$ collects deterministic components. The rank of $\Pi$ determines the number of cointegrating vectors. Under the identification of §**??**, the expected rank is $r = 1$: a single long-run relation ties output, capital, and the interaction term together, and the stationary residual of that relation is $\ln \mu_t$.

Rank determination proceeds through two complementary test sequences. The trace test evaluates $H_0 \colon r \leq j$ against $H_1 \colon r > j$ for $j = 0, 1, 2$, while the maximum-eigenvalue test evaluates $H_0 \colon r = j$ against $H_1 \colon r = j + 1$. Critical values follow **?**, adjusted for the finite-sample correction of **?** given the moderate sample lengths involved (roughly 60–70 annual observations for the US and Chile).

The deterministic specification is the restricted constant model: a drift term is absorbed into the cointegrating space rather than left unrestricted in the common trends. This choice reflects the theoretical prior that the long-run capacity relation admits a non-zero intercept — the constant $c$ in (**??**) — but that no deterministic linear trend should enter the cointegrating

residual. Intuitively, $\ln \mu_t$ fluctuates around a mean but does not trend; the restricted constant ensures that the econometric specification respects this property.

Structural-break detection employs the **?** test for cointegration with an endogenous break at an unknown date. This is particularly relevant for Chile, where the 1973 military coup and the 1982 debt crisis represent candidate regime shifts that may alter the cointegrating parameters. For the US, the candidate break corresponds to the Fordist/post-Fordist transition around 1973. The GPY (2025) impossibility result — which proves that the cointegrating vector $\beta$ cannot be fully nonparametric — provides formal justification for maintaining constant cointegrating vectors within institutional regimes and testing for breaks at regime boundaries rather than estimating continuously time-varying parameters.

## 6.2 Results: US Baseline

**[DRAFT PROSE: §2.6.2: 2–3 pages (to be filled after estimation). Report: $\hat{\beta}_1$, $\hat{\beta}_2 \Rightarrow \hat{\theta}_t^{US}$. Recovered $\widehat{\ln \mu_t^{US}}$. Regime classification: does $\hat{\theta}^{US}$ cross unity? When? Loading coefficients and speed of adjustment. Diagnostics: CUSUM, recursive eigenvalues.]**

## 6.3 Results: Chile

**[DRAFT PROSE: §2.6.3: 2–3 pages (to be filled after estimation). Same structure as §2.6.2. First structural comparison: $\hat{\theta}^{US}$ vs. $\hat{\theta}^{CL}$ — the mechanization gap in $\theta$-units. Discuss regime breaks: pre-/post-1973, pre-/post-1982.]**

## 6.4 Derived Structural Objects

Once the cointegrating parameters are recovered, the following structural objects are constructed by definition:

**Definition 6.1** (Recovered Productive Capacity)**.**

$$\hat{y}_t^p = y_t - \widehat{\ln \mu_t}. \tag{15}$$

**Definition 6.2** (Recovered Capital Productivity at Normal Capacity)**.**

$$\widehat{\ln B_t} = \hat{y}_t^p - k_t, \qquad \hat{B}_t = \exp(\hat{y}_t^p - k_t)\,. \tag{16}$$

These are the Stage B inputs: $\widehat{\mu}_t$ and $\hat{B}_t$ are passed into the behavioral accumulation law.

The recovered series $\hat{y}_t^p$ is the productive-capacity path implied by the cointegrating relation — the output level that would prevail if $\mu_t = 1$. The recovered $\hat{B}_t$ measures the efficiency of the capital stock evaluated at normal utilization, net of cyclical demand fluctuations. Its trajectory over time is the empirical counterpart of the unbalanced-growth dynamics discussed in §**??**: when $\theta(\Lambda) < 1$, $\hat{B}_t$ declines secularly as capital accumulates faster than productive capacity expands.

The two-stage architecture is load-bearing at this point. An ARDL investment equation of the type estimated in Stage B cannot perform Stage A's identification — it can describe how accumulation responds to utilization, but it cannot recover what utilization *is*. The utilization object $\widehat{\mu}_t$ and the capital-productivity object $\hat{B}_t$ are available cleanly only after the cointegrating relation has been estimated and the rank-1 identification confirmed. Section **??** derives the accounting law that decomposes accumulation into its structural components, and §**??** specifies the behavioral law that uses these recovered objects as inputs.

# 7 The Accounting Law of Capital Accumulation

This section derives the accounting law that governs the relationship between gross accumulation, profitability, capacity utilization, and capital productivity. Every step is an identity

manipulation — no behavioral equation is imposed. The behavioral content enters only through the question of whether the recapitalization rate is constant, a question deferred entirely to §**??**.

Define the gross accumulation rate as

$$\iota_t \equiv \frac{I_t}{K_t}. \tag{17}$$

The accounting decomposition of accumulation is:

$$\iota_t = \phi_t \cdot \mu_t \cdot B_t, \qquad \phi_t \equiv \frac{I_t}{Y_t}, \tag{18}$$

and since $\phi_t \cdot \mu_t = (I_t/Y_t)(Y_t/Y_t^p) = I_t/Y_t^p \equiv \phi_t^p$, we have the equivalent form

$$\iota_t = \phi_t^p \cdot B_t, \qquad \phi_t^p \equiv \frac{I_t}{Y_t^p}, \tag{19}$$

where utilization drops out at this level of abstraction. Introducing the recapitalization rate $\beta_t \equiv I_t/\Pi_t$, and using $\pi_t \equiv \Pi_t/Y_t$ and $r_t \equiv \Pi_t/K_t = \pi_t \mu_t B_t$:

$$\boxed{\iota_t = \beta_t \cdot \pi_t \cdot \mu_t \cdot B_t = \beta_t \cdot r_t.} \tag{20}$$

Equation (**??**) is an identity. There are two recapitalization lenses:

- $\beta_t = I_t/\Pi_t$: funding of investment out of profits;
- $\phi_t^p = I_t/Y_t^p$: claim of investment on productive-capacity output.

They are linked accounting representations of the same accumulation law.

Both lenses are accounting objects; neither requires a behavioral assumption. Their analytical value lies in what happens when one asks whether they are constant. If $\beta_t$ is constant, accumulation is fully determined by the accounting decomposition — capitalists mechanically reinvest a fixed share of profits regardless of structural conditions. If $\beta_t$ varies,

then the question becomes: what does it respond to? That is the behavioral margin. Note that the Weisskopf decomposition (**??**) is the profitability reading of the same identity: the profit rate is demand-led through $\mu_t$, structurally mediated through $B_t$, and distributively conditioned through $\pi_t$. Section **??** takes up the behavioral margin directly, specifying the accumulation law in which $\beta_t$ responds to distribution, demand, and the structural transformation of capital productivity.

# 8 Stage B: Behavioral Accumulation Law (ARDL)

Stage B treats the recovered structural objects $(\widehat{\mu}_t, \hat{B}_t)$ as given conditioning variables and estimates how capitalists behaviorally determine the recapitalization rate. The behavioral accumulation law is a conditional module nested inside the structural-identification architecture: it describes how realized accumulation responds to profitability once the identification problem has been solved upstream in Stage A.

## 8.1 Specification

If recapitalization responds behaviorally to distribution, demand, and structural conditions, then

$$\ln \beta_t = \eta_0 + \eta_1 \ln \pi_t + \eta_2 \ln \mu_t + \eta_3 \ln B_t, \tag{21}$$

which, substituted into the log of (**??**), yields the long-run behavioral accumulation law:

$$\boxed{\ln \iota_t = c + b_1 \ln \pi_t + b_2 \ln \mu_t + b_3 \ln B_t + \varepsilon_t} \tag{22}$$

with $b_j = 1 + \eta_j$.

**Proposition 8.1** (Dual Interpretation of Coefficients)**.** *Each coefficient in* (**??**) *has both an accounting and a behavioral reading:*

Table 2: Dual Interpretation of Accumulation Law Coefficients

| Coefficient | Accounting reading | Behavioral reading |
|---|---|---|
| $b_1$ | Profit-share elasticity of accumulation | $\eta_1 = b_1 - 1$: recapitalization response to di |
| $b_2$ | Utilization elasticity of accumulation | $\eta_2 = b_2 - 1$: recapitalization response to de |
| $b_3$ | Capital-productivity elasticity of accumulation | $\eta_3 = b_3 - 1$: recapitalization response to st |

*Notes*: Each $b_j = 1 + \eta_j$. The deviation from unity is the behavioral content. Estimation uses ARDL with Pesaran–Shin–Smith (2001) bounds test.

## 8.2 Nested Restriction Menu

The behavioral coefficients generate a hierarchy of accumulation closures:

Table 3: Nested Restriction Menu

| Restriction | Null | Model name |
|---|---|---|
| Cambridge closure | $b_1 = b_2 = b_3 = 1$ | $\beta_t$ constant |
| Bhaduri–Marglin type | $b_3 = 1$; $b_1, b_2$ free | Blind to capital productivity |
| Full structural model | All $b_j$ free | All channels active |

## 8.3 Results: US

**[DRAFT PROSE: §2.8.3: 2–3 pages (to be filled after estimation). ARDL estimation with PSS bounds test. Report $\hat{b}_1, \hat{b}_2, \hat{b}_3$ and associated Wald tests. Which closure is supported? Interpret $\eta_j = b_j - 1$ economically.]**

## 8.4 Results: Chile

**[DRAFT PROSE: §2.8.4: 2–3 pages (to be filled after estimation). Same procedure. Key question: does $b_3$ behave differently from the US? Does the capital-productivity channel operate freely or is it structurally constrained? Transition: §2.9 provides the mechanism.]**

# 9 The Peripheral Twist: BoP Constraint and the Import Content of Recapitalization

This section develops the chapter's distinctive theoretical contribution. The structural comparison of the behavioral accumulation law rests on a specific claim about why the recapitalization response differs between center and periphery. That claim originates in the work of **?** (**?**, **?**): the binding constraint on peripheral accumulation is not on output directly but on the ability to convert surplus into re-investment, because the capital goods required for accumulation must be imported.

## 9.1 The Tavares Channel

The first step of the mechanism is structural incompleteness of the capital-goods sector. Peripheral industrialization, whether under import-substitution regimes or their liberalized successors, is characterized by what **?** calls the "hard phase" of industrialization and what **?** terms "truncated industrialization": the domestic economy does not produce the full range of machinery and equipment required for capital formation. The share of machinery and equipment in gross capital formation that must be sourced from abroad — denoted $\tilde{\phi}_t^{ME}$ — is not a policy variable that governments can adjust at will but a structural parameter reflecting the depth and composition of domestic manufacturing capacity. In Latin America's larger economies, this share has oscillated historically between 0.30 and 0.60 depending on the phase of the industrialization cycle, but it has never approached zero. Even during the most ambitious import-substitution episodes, the capital-goods sector remained incomplete in the sense that the most technologically advanced vintages of equipment continued to require foreign sourcing. The consequence is that gross investment has an irreducible imported component, and the domestic availability of profits — however large — cannot by itself finance the physical act of capital formation.

The second step links this structural incompleteness to the recapitalization rate. The

accounting identity $\iota_t = \beta_t r_t$ established in §**??** decomposes the gross accumulation rate into the recapitalization rate and the profit rate. In the center, $\beta_t$ responds to the structural conditions of accumulation — to distribution, demand, and capital productivity — with relatively few binding external constraints on the conversion of profits into investment goods. In the periphery, however, the conversion channel is mediated by the foreign-exchange market. Profits exist ($\Pi_t > 0$), the profit rate may be high ($r_t$ may exceed the center's), and the desire to reinvest may be strong — yet $\beta_t$ collapses when the foreign currency required to purchase imported machinery is unavailable or prohibitively expensive. The surplus exists but cannot be converted into re-investment. This is the core of Tavares's insight, articulated most clearly in the retrospective synthesis of **?** as read through **?**: the binding constraint on peripheral accumulation is not the generation of surplus but its physical realization as installed productive capital.

The third step identifies the historical transmission mechanism. The constraint on $\beta_t$ does not operate continuously but activates at discrete episodes of external financial stress — debt-cycle reversals, terms-of-trade collapses, sudden stops in capital inflows. The Volcker shock of 1979–1982 is the canonical example for Latin America: the sudden repricing of dollar-denominated debt simultaneously drained foreign-exchange reserves and raised the domestic-currency cost of imported capital goods, producing a collapse in $\beta_t$ that persisted for nearly a decade despite the partial recovery of domestic profitability. These episodes express financial dependency as a constraint on the technical composition of accumulated capital. They are not generic recessions in which output and investment fall together; they are structurally asymmetric crises in which the capacity to generate surplus and the capacity to convert that surplus into productive capital are severed by the foreign-exchange constraint.

It is essential to distinguish the Tavares channel from two superficially similar but analytically distinct mechanisms. It is not a pure output-side balance-of-payments constraint of the **?** variety, in which export growth determines the growth of output through the income elasticity of imports — although the two may operate simultaneously. And it is not simple

import compression, in which a foreign-exchange shortage reduces the volume of all imports proportionally. The Tavares channel is specifically a recapitalization constraint: it operates on $\beta_t$, not on $Y_t$, and it binds through the import content of investment goods, not through the aggregate import bill. The contribution of this chapter is to give that theoretical distinction empirical content by estimating the behavioral accumulation law separately for the US and Chile and testing whether the coefficient $b_3$ — the recapitalization response to capital productivity — is structurally constrained in the periphery relative to the center.

## 9.2 Formalization

The Tavares channel described in §**??** identifies a structural constraint on $\beta_t$ in the periphery, operating through the import content of investment and the availability of foreign exchange. The key claim is that the recapitalization response to capital productivity — the parameter $\eta_3 = b_3 - 1$ in the behavioral accumulation law (**??**) — is not a free behavioral parameter in the periphery. It is structurally constrained by two forces: the import content of investment $\tilde{\phi}_t^{ME}$ and the external financial conditions summarized by a balance-of-payments proxy $\mathrm{BoP}_t$.

Disaggregated import data for Chile at the machinery-and-equipment level are not available at annual frequency over the full sample, so the empirical strategy adopts the **?** prior: from the cross-country tabulation in Cuadro 8, the machinery-and-equipment share of capital formation in semi-industrialized economies clusters around $\xi_K^{ME} \approx 0.92$–$0.94$. This value is imposed as an empirical prior, not estimated from unavailable disaggregated data. The balance-of-payments proxy $\mathrm{BoP}_t$ admits several candidate series — the terms of trade, the real exchange rate, or the current-account balance — and the choice among them is an empirical question adjudicated by the Stage B estimation.

Three candidate specifications implement the constraint. The first augments the Chilean ARDL with a foreign-exchange availability proxy, entering $\mathrm{BoP}_t$ directly as an additional regressor in the behavioral accumulation law and testing whether it absorbs part of the $b_3$

channel. The second tests whether $b_3^{CL}$ is stable across balance-of-payments regimes or breaks at sudden-stop episodes — the preferred implementation is the **?** threshold VECM, in which the balance-of-payments constraint activates as a threshold on $\tilde{\phi}_t^{ME}$ rather than operating linearly throughout the sample. The third compares the unconstrained US estimate $b_3^{US}$ directly against the Chilean estimate $b_3^{CL}$ and tests the structural null $\eta_3^{US} > \eta_3^{CL}$: the peripheral recapitalization response to capital productivity is more constrained than the center's.

It is important to distinguish two sources of external constraint that the literature sometimes conflates. The income elasticity of exports $\varepsilon$ is an exogenous parameter reflecting the composition of global trade. The import propensity is endogenous to the domestic productive structure: it reflects the degree of structural incompleteness of the capital-goods sector. The Kaldor-versus-ECLA fault line turns on whether the external ceiling on accumulation is determined primarily by $\varepsilon$ (the export side, as in Thirlwall) or by the import propensity operating through $\tilde{\phi}_t^{ME}$ (the import-content side, as in Tavares). This is an estimable structural question, and the threshold-VECM specification is designed to adjudicate it.

Let $\tilde{\phi}_t^{ME}$ denote the import content of investment — the share of machinery and equipment in gross capital formation that is sourced from abroad. The peripheral behavioral law gains a forex-mediation channel:

$$\eta_3^{CL} = \eta_3^{CL}\left(\tilde{\phi}_t^{ME}, \mathrm{BoP}_t\right), \tag{23}$$

where $\mathrm{BoP}_t$ is a measure of external constraint intensity (terms of trade, real exchange rate, or current account balance).

## 9.3 The Technical Composition Channel

The Tavares channel operates through the recapitalization rate; the technical composition channel explains why over-mechanization amplifies the constraint. When the capacity trans-

formation elasticity satisfies $\theta < 1$ — the over-mechanization regime identified in §**??** — two consequences follow simultaneously, and their interaction produces a structural trap specific to the periphery.

The first consequence is that more capital is required per unit of productive capacity created. Under over-mechanization, the capital stock grows faster than productive capacity: $\ln B_t = (\theta - 1)k_t < 0$ when $\theta < 1$, so capital productivity at normal capacity falls secularly. Each successive unit of capacity requires a larger capital outlay. In an economy with a complete domestic capital-goods sector, this is a drag on accumulation but not a binding constraint — the additional capital goods can be produced domestically, financed from the surplus, and installed without crossing the foreign-exchange market. In the periphery, however, the structural incompleteness described in §**??** means that the additional capital goods must be imported. Over-mechanization does not merely raise the volume of investment required; it raises the volume of *imported* investment required, because the marginal capital goods needed to sustain capacity growth are disproportionately those that the domestic economy does not produce.

The second consequence is that the economy's capacity to earn the foreign exchange needed to finance those imports deteriorates. If capital productivity falls, the output obtainable from the installed capital stock at normal utilization declines, and with it the export base that generates foreign-currency earnings. The secular deterioration of the terms of trade identified by Prebisch and Singer is not, in this framework, merely a price phenomenon — it is a structural constraint on the technical composition of peripheral capital accumulation. The economy needs more imported machinery per unit of capacity, but its capacity to earn the forex that pays for that machinery is lower because the accumulated capital is less productive.

The peripheral trap is the interaction of these two forces. Over-mechanization raises the import bill for investment while degrading the export capacity that finances it. The $b_3$ comparison in §**??** gives this trap empirical content: if $b_3^{CL} < b_3^{US}$, then Chilean capitalists can-

not recapitalize at the rate the structural conditions demand, because the foreign-exchange constraint binds. The finding would indicate that the recapitalization response to capital productivity is structurally dampened in the periphery — not because Chilean capitalists lack the will or the surplus to reinvest, but because the physical conversion of surplus into installed productive capital requires imported machinery that the balance of payments cannot finance.

This is the chapter's central structural claim about the center–periphery difference: *demand conditions intervene on the political economy of the ceiling, i.e. the class struggle over distribution, utilization, and the institutional forms that govern whether accumulation can proceed.*

# 10 Structural Comparison: Center vs. Periphery

**[DRAFT PROSE: §2.10 intro: 1 paragraph. Synthesize the results across both countries. The parallel construction (§2.5.3) ensures the comparison is structural, not an artefact of measurement.]**

Table 4: Center–Periphery Structural Comparison

| Diagnostic | US | Chile | Reading |
|---|---|---|---|
| $\hat{\theta}$ | — | — | Mechanization gap |
| Regime breaks | — | — | When does $\theta$ shift? |
| $b_1$ (distribution) | — | — | Profit-share response |
| $b_2$ (demand) | — | — | Utilization response |
| $b_3$ (structural) | — | — | Capital-productivity response |
| $b_3$ stability | — | — | Does $b_3^{CL}$ break at BoP crises? |
| $\widehat{\mu}_t$ volatility | — | — | Utilization instability |
| $\hat{B}_t$ trend | — | — | Capital productivity trajectory |

*Notes*: Cells filled after Stage A and Stage B estimation. $b_3$ is the key comparison: in the US, the recapitalization function responds to the capital-productivity channel relatively freely; in Chile, $b_3^{CL}$ is mediated by the BoP constraint (§2.9).

**[DRAFT PROSE: §2.10 narrative: explain each row of Table ??. Punchline:**

financial dependency shows up empirically as a structurally constrained recapitalization response. The Tavares channel provides the mechanism; Table ?? provides the evidence.]

## 11 Conclusions

[DRAFT PROSE: §2.11: approx. 3 pages. Three deliverables: (1) CU series: structurally identified $\hat{\mu}_t$ for both countries, available as inputs to Ch3. (2) Regime classification: $\hat{\theta}^{US}$ vs. $\hat{\theta}^{CL}$ — the empirical realization of the core-periphery mechanization gap. (3) Financial dependency formalized: the Tavares channel — inability to convert surplus into re-investment due to lack of foreign currency — is not a narrative but an estimable structural constraint on the recapitalization function. Forward references: Ch3 inherits $\hat{\mu}_t$, $\hat{B}_t$, $\hat{\theta}_t$ as given inputs for profitability analysis and investment function hypothesis testing. The Weisskopf decomposition and Okishio crisis trigger become Ch3 applications of the Stage A recovered objects.]